SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 3 – Error Analysis Task
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	D/1st year	**Date:**	08-10-2026

Code of Conduct

I G.SANGHAMITRA,(1924260351) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G Sanghamitra

Instructions

- Read each question carefully; answer all four questions (100 in total). Off-Class Assessment Tool.
- Each question gives a student's faulty work. Do not solve from scratch: identify every error (cite it by statement, component, iteration or step number), explain its root cause, correct it with justification and apply the corrected concept. An error merely carried forward counts once.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Error Analysis rubric: error identification, root cause analysis, correction and justification, application of concepts, and clarity and presentation.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	**Q1**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**

7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	**Q3**

9. Uninformed Search Strategies	Error tracing of BFS, DFS and uniform-cost search	**Q3**
10. Search with Partial Information	Sensorless problems, belief states, percept update	**Q4**

Question 1: Foundations of AI, Intelligent Agents and Rationality – Error Analysis of Concept Notes

Topic: Foundations and history of AI, agents and environments, rationality, nature of environments, structure of agents

A junior engineer prepared the revision notes below on the foundations of AI and intelligent agents. The notes contain errors; some statements may be correct. Analyse them statement by statement.

No.	Statement in the notes
S1	The Turing Test (1950) judges a machine to be intelligent if its internal reasoning always follows the laws of logic. It is therefore the standard example of the "thinking rationally" approach to AI.
S2	The "AI winters", periods of reduced funding and interest, occurred in the 1950s, before the 1956 Dartmouth workshop. That workshop then revived the field and coined the term "artificial intelligence".
S3	A rational agent always achieves the best possible outcome because it knows the actual result of every action in advance. Hence a taxi that picks a route later blocked by an unforeseeable landslide was not rational.
S4	PEAS description of an automated taxi – Performance measure: steering, accelerator, brake, horn. Environment: roads, other traffic, pedestrians, weather. Actuators: safe, fast, legal and comfortable trip. Sensors: cameras, GPS, speedometer.
S5	Chess played without a clock is a partially observable, stochastic, episodic, static and discrete task environment.
S6	A model-based reflex agent keeps an internal state, updated using knowledge of how the world evolves and how its own actions affect it. It can therefore act sensibly even when the current percept does not reveal the full state of the world.

(a) Identify which statements are correct and which are incorrect. For each incorrect statement, quote the erroneous phrase or claim (one statement contains more than one error).

Statement	Correct / Incorrect	Erroneous phrase / claim
S1		
S2		
S3		
S4		
S5		
S6		

(b) Root cause analysis: for every error, explain the underlying misconception, i.e. which concept has been misunderstood and why the statement goes wrong.

Error (statement / part)	Root cause (misconception)

(c) Rewrite each faulty statement correctly and justify your correction. For a statement that is already correct, write "No change" and justify why.

Statement	Corrected statement	Justification
S1		
S2		
S3		
S4		
S5		
S6		

(d) Apply the corrected concepts to the following case. A self-driving taxi chooses a highway route that every available traffic report showed to be clear; an unforeseeable landslide then blocks the road.

(i) Using the four factors that decide whether an agent is rational, argue whether the taxi's choice was rational.

(ii) Name the properties of the taxi's task environment that make perfect foresight impossible.

(iii) Describe one situation in which a simple reflex taxi would behave poorly, and state what a model-based reflex taxi keeps in order to avoid it.

(i)		
(ii)		

(iii)	

Question 2: Problem-Solving Agents and Problem Formulation – Error Analysis of a Formulation

Topic: Problem-solving agents, problem formulation

A student formulates the **Missionaries and Cannibals** problem for a problem-solving agent. Three missionaries and three cannibals stand on the left bank of a river, with a boat that carries one or two people and cannot cross empty. The boat is on the left bank at the start. Missionaries must never be outnumbered by cannibals on either bank (a bank with no missionaries is safe). The objective is to move everyone to the right bank using the **fewest crossings**.

Component	Student's formulation
State representation	(M, C) = number of missionaries and cannibals on the left bank
Initial state	(3, 3)
Actions	Take the boat across with 1M, 2M, 1C, 2C or 1M + 1C
Goal test	M = C
Legality check	A state is legal if C ≤ M on the left bank (or M = 0 there)
Path cost	Number of people carried in the crossing
Note N1	The agent works in the order: search → goal formulation → problem formulation → execution.
Note N2	The state space contains 16 states, of which 10 are legal.

(a) For each component and note, state whether it is correct or incorrect; for each incorrect one, state what is wrong.

Component / Note	Correct / Incorrect	What is wrong
State representation		
Initial state		
Actions		
Goal test		
Legality check		
Path cost		
Note N1		
Note N2		

(b) Root cause analysis: for every error, explain the underlying cause, i.e. which principle of problem formulation or of the problem-solving agent has been violated.

Error (component / note)	Root cause

(c) Give the corrected version of every faulty component or note and justify each correction.

Component / Note	Corrected formulation	Justification

(d) Apply your corrected formulation.

(i) Complete the table of boat loads available from the initial state, writing each resulting state as (M, C, boat side) and whether it is legal, with the reason.

Boat load	Resulting state (M, C, boat)	Legal? Reason
1M		
2M		
1C		
2C		
1M + 1C		

(ii) Give one state that the student's legality check accepts although it is actually illegal.

(iii) Using your corrected path cost, state the cost of an optimal solution.

(ii)	
(iii)	

Question 3: Searching for Solutions and Uninformed Search – Error Analysis of Search Traces

Topic: Searching for solutions, BFS, DFS, uniform-cost search

A student ran BFS, DFS and uniform-cost search (UCS) on the weighted undirected graph below. The start node is S and the goal node is G. **Conventions:** graph search with an explored set; neighbours expanded alphabetically; goal test when a node leaves the frontier; ties broken alphabetically. BFS and DFS ignore edge costs while searching, but path costs are measured with them. **BFS:** do not add a node that is already explored or already in the queue. **DFS:** use a stack with its top on the left; push children so that the alphabetically first child is popped first; do not push a node that is already explored or already on the stack. **UCS:** write entries as Node:g(parent) and replace an entry whenever a cheaper path to a frontier node is found.

Edge	Cost	Edge	Cost
S – C	3	B – G	3
S – D	3	D – E	2
A – B	2	D – G	6
A – C	6	E – G	3
B – D	4		

Student's BFS trace (queue initially [S])

Iteration	Node Dequeued	Queue after Expansion	Explored Set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[B, E, G, A]	{S, C, D}
4	B	[E, G, A]	{S, C, D, B}
5	E	[G, A]	{S, C, D, B, E}
6	G (goal)	[A]	{S, C, D, B, E}

BFS result: path S – D – G, cost 9, 6 nodes removed from the frontier.

Student's DFS trace (stack initially [S], top on the left)

Iteration	Node Popped	Stack after Pushing Children
1	S	[D, C]
2	D	[G, E, B, C]
3	G (goal)	[E, B, C]

DFS result: path S – D – G, cost 9, 3 nodes removed from the frontier.

Student's UCS trace

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), A:9(C), G:9(D)
5	B	7	A:9(C), G:9(D)
6	A	9	G:9(D)
7	G (goal)	9	–

UCS result: path S – D – G, cost 9, 7 nodes removed from the frontier.

Student's remarks

No.	Remark
R1	An explored set is not needed in BFS because BFS never revisits a node, so the Explored Set column is unnecessary.
R2	All three strategies returned S – D – G with cost 9, so all three are optimal and the traces are confirmed correct.
R3	DFS removed the fewest nodes from the frontier (3), so DFS is the best strategy for this graph.

(a) Examine the three traces and the remarks. List every error, giving the trace and iteration (or the remark number) and what is wrong. An error that is only carried forward counts once.

No.	Where (trace / iteration / remark)	What is wrong
1		
2		
3		
4		
5		
6		
7		

(b) Root cause analysis: for each error listed in (a), explain the rule or concept that was violated and how it led to the wrong entry or conclusion.

No.	Root cause (rule or concept violated)
1	

2	
3	
4	
5	
6	
7	

(c) Write the corrected traces and justify each correction.

(i) BFS: the corrected queue after iteration 3 and the full order in which nodes are dequeued.

Queue after iteration 3	
Order of dequeued nodes	

(ii) DFS: the corrected trace.

Iteration	Node Popped	Stack after Pushing Children
1		
2		
3		
4		
5		

(iii) UCS: the corrected trace.

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1			
2			
3			
4			
5			
6			
Justification of the corrections			

(d) Apply the corrected traces.

(i) Complete the comparison table.

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?

BFS				
DFS				
UCS				

(ii) Use iteration 3 of the corrected BFS trace to explain why an explored set is needed in graph search.

(ii)	

Question 4: Search with Partial Information – Error Analysis of BeliefState Computation

Topic: Problem formulation, search with partial information (sensorless and percept-based belief-state search)

A vacuum agent lives in a two-square world (Left square, Right square), each square being Clean (C) or Dirty (D). Actions: Left, Right and Suck (Left in the left square and Right in the right square leave the agent where it is; Suck cleans the current square). The goal is that both squares are Clean. Physical states are written (Agent, Left square, Right square):

State	Description	State	Description
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

Task A – sensorless agent. The agent knows it is in the Left square but nothing about the dirt, so its initial belief state is {S1, S3, S5, S7}. A student applied the plan [Suck, Right, Suck]:

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S6, S8}

Task B – agent with a local sensor that reports [location, status of the current square]. Starting from the belief state of all eight states, the agent receives the percept [Left, Dirty], performs Suck, receives [Left, Clean], performs Right and receives [Right, Dirty]. The student's table:

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S2, S3, S6}
1	Suck	{S4, S5, S7, S8}	[Left, Clean]	{S5, S7}
2	Right	{S6, S7}	[Right, Dirty]	{S6}

Student's conclusions

No.	Conclusion
C1	Task A: the final belief state still contains two states, so the agent can never know where it is. No plan can guarantee the goal when the agent has no sensors; hence sensorless problems cannot be solved.
C2	Task B: the belief state is now the singleton {S6}, so the agent knows its location and the dirt status, and the goal has been reached.

C3	With 8 physical states there are only 8 belief states (one per physical state), so searching over belief states is no harder than searching over physical states.

(a) Examine the tables and the conclusions. List every error, giving the task and step (or the conclusion number) and what is wrong. An error that is only carried forward from an earlier wrong row counts once.

No.	Where (task / step / conclusion)	What is wrong
1		
2		
3		
4		
5		
6		
7		

(b) Root cause analysis: for each error listed in (a), explain the concept or rule of belief-state search that was misapplied.

No.	Root cause (concept or rule misapplied)
1	
2	
3	
4	
5	
6	
7	

(c) Write the corrected tables and conclusions, with justification.

(i) Task A: the corrected belief states.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	

2	Right	
3	Suck	

(ii) Task B: the corrected belief states after each action and each percept.

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	
1	Suck		[Left, Clean]	
2	Right		[Right, Dirty]	

(iii) Corrected conclusions C1 and C2, with justification.

C1	
C2	

(d) Apply the corrected concepts.

(i) How many different belief states are possible when there are 8 physical states? Why does this make belief-state search harder than search over physical states?

(i)	

(ii) For the initial belief state {S1, S3, S5, S7}, apply the plan [Right, Suck], write the belief state after each action, and state whether it is a solution.

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Right	
2	Suck	
	Solution? Why?	

Rubrics for Error Analysis

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Error Identification	**20**	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors

Root Cause Analysis	**30**	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	**25**	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections are partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	**15**	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	**10**	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria	Error Identification	Root Cause Analysis	Correction & Justification	Application of Concepts	Clarity & Presentation	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

SOLUTIONS

Question 1 – Foundations of AI, Intelligent Agents and Rationality

(a) Correct / Incorrect

Statement	Correct / Incorrect	Erroneous phrase / claim
S1	Incorrect	"judges a machine to be intelligent if its internal reasoning always follows the laws of logic" and "standard example of the thinking rationally approach"
S2	Incorrect	"AI winters … occurred in the 1950s, before the 1956 Dartmouth workshop" and "That workshop then revived the field"
S3	Incorrect	"always achieves the best possible outcome because it knows the actual result of every action in advance" and "was not rational"
S4	Incorrect	Performance measure and actuators are swapped: "Performance measure: steering, accelerator, brake, horn" and "Actuators: safe, fast, legal and comfortable trip"
S5	Incorrect	"stochastic" and "episodic" are incorrect; chess is deterministic and sequential
S6	Correct	No error

(b) Root Cause Analysis

Error	Root cause / misconception
S1 – Turing Test is based on logical reasoning	The Turing Test evaluates whether a machine can behave like a human in conversation, not whether its internal reasoning follows formal logic. It belongs to the acting humanly approach.
S1 – Thinking rationally approach	Thinking rationally is associated with correct logical reasoning. The Turing Test instead focuses on externally observable intelligent behaviour.
S2 – AI winters occurred in the 1950s before Dartmouth	AI winters occurred later, notably in the 1970s and late 1980s–early 1990s, when funding and interest declined.
S2 – Dartmouth workshop revived the field	The 1956 Dartmouth workshop was a foundational event for AI, not a response to an earlier AI winter.
S3 – Rational agent knows every result in advance	Rationality does not require perfect knowledge or foresight. The agent chooses the action expected to produce the best outcome using its percept history, knowledge and available actions.
S3 – Landslide makes taxi irrational	An unforeseeable event does not make a previously rational decision irrational. The decision is judged using information available when the action was selected.
S4 – Performance measure = steering, brake, horn	These are actuators because they are mechanisms through which the taxi acts on the environment.
S4 – Actuators = safe, fast, legal, comfortable	These are desired qualities used to evaluate the taxi, so they belong to the performance measure.
S5 – Chess is stochastic	Standard chess is deterministic; there is no random element in legal moves or game rules.
S5 – Chess is episodic	Chess is sequential because every move affects future moves and the eventual result.
S6	No misconception; the statement correctly describes a model-based reflex agent.

(c) Corrected Statements and Justification

Statement	Corrected statement	Justification
S1	The Turing Test (1950) evaluates whether a machine can exhibit human-like behaviour in conversation. It is a	The Turing Test focuses on observable behaviour rather than

	standard example of the "acting humanly" approach to AI, not the "thinking rationally" approach.	checking whether internal reasoning follows formal logic.
S2	AI winters were periods of reduced funding and interest that occurred after the early development of AI, notably in the 1970s and late 1980s–early 1990s. The 1956 Dartmouth workshop was a foundational event in AI and is associated with the introduction of the term "artificial intelligence".	The AI winters did not occur before Dartmouth. Dartmouth helped establish AI as a field.
S3	A rational agent chooses the action expected to maximize its performance measure based on its percept history, knowledge and available actions. It does not need to know the actual result of every action in advance. A taxi choosing a route later blocked by an unforeseeable landslide can still be rational.	Rationality is based on the information available when the decision is made, not on perfect hindsight.
S4	Performance measure – safe, fast, legal and comfortable trip. Environment – roads, other traffic, pedestrians and weather. Actuators – steering, accelerator, brake and horn. Sensors – cameras, GPS and speedometer.	PEAS means Performance measure, Environment, Actuators and Sensors.
S5	Chess played without a clock is a partially observable, deterministic, sequential, static and discrete task environment.	Chess has no randomness, each move affects future states, the board is static while a player thinks, and states/actions are discrete.
S6	No change.	A model-based reflex agent maintains an internal state using knowledge of how the world changes and how its actions affect the world.

(d) Application

(i) Rationality

Yes, the taxi's choice was rational.

- Performance measure: The taxi should provide a safe, fast, legal and comfortable journey.
- Percept sequence: At the time of choosing the route, all available traffic reports indicated that the highway was clear.
- Knowledge of the environment: The taxi had no information indicating that a landslide would occur.
- Available actions: Choosing the clear highway was a reasonable action among the available routes.

Therefore, the taxi made the best expected decision using the information available at that time. Since the landslide was unforeseeable, the later bad outcome does not mean the original decision was irrational.

(ii) Task-environment properties

- Partially observable – the taxi cannot observe every condition everywhere.
- Stochastic – unexpected events such as accidents, weather changes or landslides can occur.
- Dynamic – the environment can change while the taxi is operating.
- Sequential – current actions affect future states and decisions.
- Uncertain/unknown – the taxi may not know every future event or environmental condition.

(iii) Simple reflex vs model-based reflex

A simple reflex taxi may behave poorly when the current percept is insufficient. For example, it may see that the road directly ahead is clear and continue forward even though the road is closed further ahead.

A model-based reflex taxi keeps an internal state and a model of how the world changes and how its actions affect the world. This helps it make sensible decisions when the current percept does not reveal the complete situation.

Question 2 – Problem-Solving Agents and Problem Formulation

(a) Correct / Incorrect

Component / Note	Correct / Incorrect	What is wrong
State representation	Incorrect	The boat side is missing. A complete state should include missionaries, cannibals and boat side, e.g. (M, C, B).
Initial state	Incorrect	It should be (3, 3, Left), not just (3, 3).

Actions	Correct	The listed boat loads are the possible loads; legality is checked from the resulting state.
Goal test	Incorrect	The goal is everyone on the right bank: M = 0 and C = 0 on the left (with the boat side included in the state).
Legality check	Incorrect	Both banks must satisfy the missionaries-not-outnumbered condition, not just the left bank.
Path cost	Incorrect	The objective is the fewest crossings, so each crossing should have cost 1.
Note N1	Incorrect	The correct order is goal formulation → problem formulation → search → execution.
Note N2	Incorrect	With the boat side included there are 32 possible physical states; 20 are legal. The student's 16/10 count omits the boat side.

(b) Root Cause Analysis

Error	**Root cause**
State representation	The student treated the boat position as irrelevant, but the next legal action depends on which bank contains the boat.
Initial state	The boat's initial location is part of the state and must be represented.
Goal test	The student confused equality of left-bank counts with the actual goal of moving everyone to the right bank.
Legality check	Safety must be checked on both banks because cannibals must not outnumber missionaries on either bank.
Path cost	The problem explicitly asks for the fewest crossings, so crossing count—not people carried—is the path cost.
Note N1	Problem solving starts by defining the goal and then formulating the problem before searching and executing.
Note N2	Omitting the boat side undercounts the state space. Including it doubles the 16 M/C combinations to 32 physical states; 20 are legal.

(c) Corrected Formulations

Component / Note	**Corrected formulation**	**Justification**
State representation	(M, C, B), where M and C are the numbers on the left bank and B is the boat side (L/R).	The boat side determines which people can move next.
Initial state	(3, 3, L).	All six people and the boat start on the left bank.
Actions	Move the boat with 1M, 2M, 1C, 2C, or 1M+1C, provided the resulting state is legal.	These are the permitted boat loads; illegal resulting states are rejected.
Goal test	M = 0 and C = 0 (everyone is on the right bank).	This directly represents the objective.
Legality check	Both banks must satisfy the safety condition: missionaries are not outnumbered by cannibals unless there are no missionaries on that bank.	Safety applies to both banks.
Path cost	1 per crossing; total path cost = number of crossings.	The objective is the fewest crossings.

Note N1	Goal formulation → problem formulation → search → execution.	The goal is defined before the detailed problem and search.
Note N2	There are 32 possible states including boat side; 20 are legal.	There are 16 M/C combinations × 2 boat positions, with 20 satisfying both-bank safety.

(d) Application

(i) Boat loads from the initial state

Boat load	Resulting state (M, C, boat)	Legal? Reason
1M	(2, 3, R)	No – left bank has 2 missionaries and 3 cannibals.
2M	(1, 3, R)	No – left bank has 1 missionary and 3 cannibals.
1C	(3, 2, R)	Yes – both banks satisfy the safety condition.
2C	(3, 1, R)	Yes – both banks satisfy the safety condition.
1M + 1C	(2, 2, R)	Yes – both banks satisfy the safety condition.

(ii)

Example: (2, 3, R). The student's check accepts it because it only checks the left-bank condition incorrectly; in reality, the left bank has more cannibals than missionaries, so it is illegal.

(iii)

Optimal path cost = 11 crossings.

Question 3 – Searching for Solutions and Uninformed Search

(a) Errors in the traces and remarks

No.	Where	What is wrong
1	BFS – Iteration 3	After expanding D, the existing frontier node A must remain before D's newly generated nodes. Correct queue: [A, B, E, G], not [B, E, G, A].
2	DFS – Iteration 1	The stack is ordered incorrectly. With the stated convention, C must be popped before D, so the stack should be [C, D], not [D, C].
3	UCS – Iteration 4	Expanding E with cost 5 gives a cheaper route to G: S-D-E-G has cost 8, replacing G:9(D) with G:8(E).
4	R1	An explored set is needed in graph search because graphs can contain repeated/revisited nodes.
5	R2	The three strategies are not all optimal. BFS and DFS are not guaranteed to minimize weighted path cost; UCS is optimal here.
6	R3	Removing fewer nodes does not make DFS the best strategy. Here DFS finds a higher-cost path, S-C-A-B-G, with cost 14.

(b) Root Cause Analysis

No.	Root cause
1	The BFS frontier order was not preserved. Existing frontier nodes must remain ahead of newly appended children.
2	The DFS stack convention was misread. Since the top is on the left and the alphabetically first child must be popped first, C must be placed at the left.
3	UCS must replace a frontier entry whenever a cheaper path to that same frontier node is discovered.
4	Graph search needs an explored set to prevent repeated expansion and cycles.
5	BFS and DFS ignore edge costs when searching; UCS considers path cost and is the appropriate optimal strategy for positive edge costs.
6	Efficiency cannot be judged only by the number of nodes removed. Solution quality, optimality and completeness also matter.

c) Corrected Traces

(i) BFS

Queue after iteration 3: [A, B, E, G]

Order of dequeued nodes: S, C, D, A, B, E, G

(ii) DFS

Iteration	Node Popped	Stack after Pushing Children
1	S	[C, D]
2	C	[A, D]
3	A	[B, D]
4	B	[G, D]
5	G (goal)	[D]

(iii) UCS

Iteration	Node Expanded	g(n)	Priority Queue after Expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), G:8(E), A:9(C)
5	B	7	G:8(E), A:9(C)
6	G (goal)	8	–

Justification: BFS preserves the existing frontier order; DFS follows the specified alphabetical stack rule; UCS replaces G:9(D) with the cheaper G:8(E) and then removes G at minimum cost.

(d) Application

Strategy	Path Found	Path Cost	Nodes Removed from Frontier	Optimal Here?
BFS	S–D–G	9	7	No

DFS	S–C–A–B–G	14	5	No
UCS	S–D–E–G	8	6	Yes

Iteration 3 of corrected BFS shows why an explored set is needed: when D is expanded, some neighbours are already explored or already in the frontier. The explored set prevents repeated expansion and graph cycles.

Question 4 – Search with Partial Information and Belief States

(a) Errors

No.	Where	What is wrong
1	Task A – Step 3	After Suck, the belief state is {S8}, not {S6, S8}. Suck cleans the right square in both possible states.
2	Task A – C1	The final belief state does not need to be a singleton to guarantee the goal. {S8} is actually a singleton, and the plan achieves both squares clean.
3	Task B – Step 0	After percept [Left, Dirty], the belief state is {S1, S3}, not {S1, S2, S3, S6}.
4	Task B – Step 1	The predicted state after Suck should be {S5, S7}, not {S4, S5, S7, S8}.
5	Task B – Step 2	After Right, the predicted belief state should be {S6, S8}, not {S6, S7}.
6	Task B – percept update / C2	After [Right, Dirty], the belief state is {S6}; therefore the right square is dirty and the goal has not been reached.
7	C3	With 8 physical states, there are 2^8 = 256 possible belief states (including the empty subset), not only 8.

(b) Root Cause Analysis

No.	Root cause
1	The student did not apply the Suck action to every state in the belief state correctly.
2	The student confused knowing the exact physical state with guaranteeing the goal. A belief state can contain multiple states and still guarantee a goal if every state in it satisfies the goal.

3	A percept filters the current belief state to states consistent with the observation. [Left, Dirty] leaves only S1 and S3.
4	The action update must be applied to every state in the current belief state. Suck changes S1→S5 and S3→S7.
5	Right changes S5→S6 and S7→S8, so the predicted belief state is {S6, S8}.
6	The percept [Right, Dirty] filters {S6, S8} to {S6}; S6 still has the right square dirty, so the goal is false.
7	A belief state is a set of possible physical states. For n physical states, there are 2^n possible subsets.

(c) Corrected Tables and Conclusions

(i) Task A

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Suck	{S5, S7}
2	Right	{S6, S8}
3	Suck	{S8}

(ii) Task B

Step	Action	After Action (Predicted)	Percept	After Percept (Updated)
0	(start)	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S3}
1	Suck	{S5, S7}	[Left, Clean]	{S5, S7}
2	Right	{S6, S8}	[Right, Dirty]	{S6}

(iii) Corrected conclusions

C1: The plan [Suck, Right, Suck] guarantees the goal. The final belief state is {S8}; both squares are clean. The agent can solve a sensorless problem without necessarily knowing the exact state at every step.

C2: The final belief state is {S6}. The agent knows the physical state, but the goal has not been reached because the right square is still dirty.

(d) Application

(i)

There are 2^8 = 256 possible belief states (including the empty set). Belief-state search can therefore be much larger than search over the 8 physical states because each search state is a subset of physical states.

(ii)

Step	Action	Belief State after Action
0	(start)	{S1, S3, S5, S7}
1	Right	{S2, S4, S6, S8}
2	Suck	{S4, S8}